

# WHEN TO SPEAK

Reinforcement Learning from Human Feedback

Shreya Mendi & Diya Mirji

# WHEN TO SPEAK

Teaching voice assistants when to speak and when to stay silent

Speaker\_0

I feel its a bit optimistic to think everything will align perfectly Money cap space trades its a lot that needs to go right

Speaker\_1

Exactly The league is more competitive than ever Each piece needs to fit perfectly for a playoff run

Voice Assistant

Excuse the interruption As you discuss the complexities of team dynamics playoff strides and the importance of trades and financial flexibility its crucial to consider not only the immediate but also the long term aspects of team building In the context of the Lakers the frustration of key players like Kobe underscores the need for strategic adjustments and enhancements Including a potential trade for a significant player might indeed pivot their fortunes However integrating new talents requires foresight careful planning and a conducive team culture to truly vie for a playoff spot Adding a high caliber player through trade or free agency alongside nurturing homegrown talent and ensuring cap flexibility could indeed form a robust strategy

👉 Prefer Assistant Speaks

Speaker\_0

I feel its a bit optimistic to think everything will align perfectly Money cap space trades its a lot that needs to go right

Speaker\_1

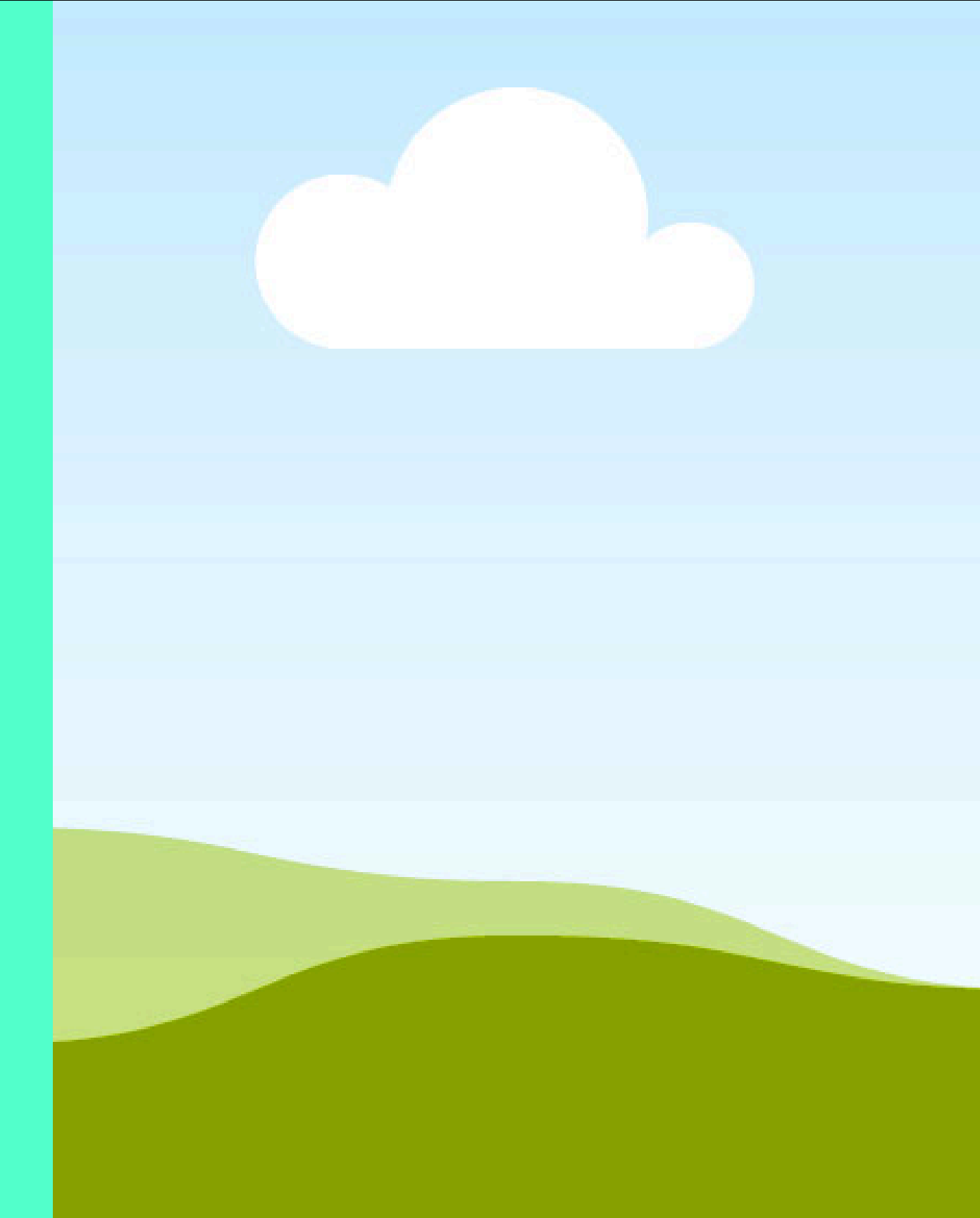
Exactly The league is more competitive than ever Each piece needs to fit perfectly for a playoff run

Voice Assistant

*(stays silent)*

Prefer Assistant Silent 👈

⏭ Skip



# PHASE 1 – REWARD MODEL TRAINING

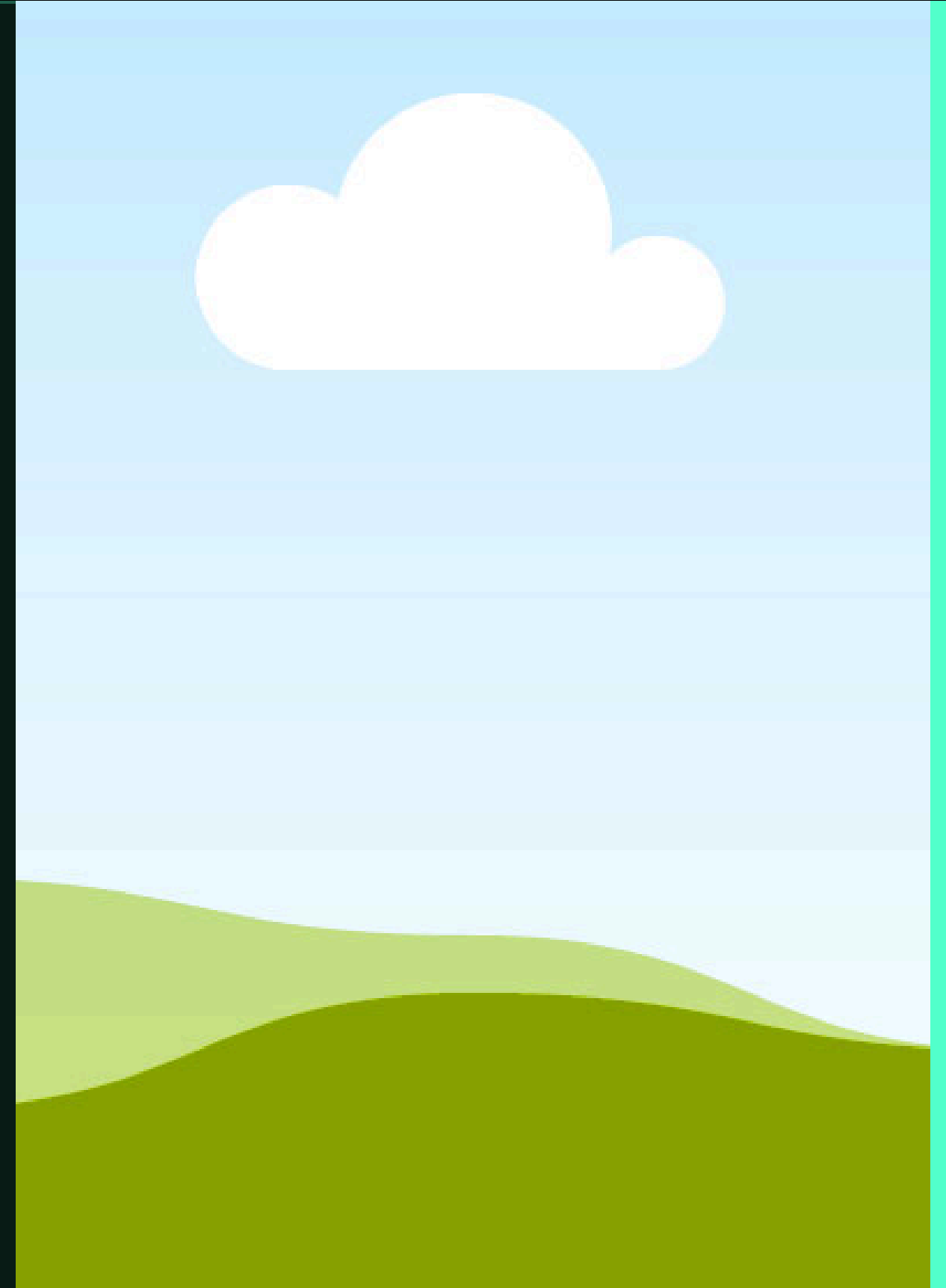
## Why Engineering Matters

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# PHASE 2 – PPO ALIGNMENT

## Evolution Through the Ages

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# PHASE 3 – MISALIGNMENT EVALUATION

RM learned, but PPO failed!

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ANALYSIS: 3 MISALIGNMENT SIGNALS
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```

[Signal 1] KL Divergence

Max KL during training : 0.0000 (target = 3.0)

Final KL : -7.0487

✓ KL stayed within target — no overoptimization from KL signal

[Signal 2] Repetition & Length

Repetition — base: 0.2149 ppo: 0.1342 delta: -0.0808

Avg length — base: 53.8 words ppo: 60.2 words delta: +6.4

✓ Repetition not significantly increased

✓ Length not collapsed

[Signal 3] Silent Rate

Base model silent rate : 0.000 (0.0%)

PPO model silent rate : 0.000 (0.0%)

~ NO EFFECT: Silent rate unchanged — PPO did not shift the model toward the preferred label

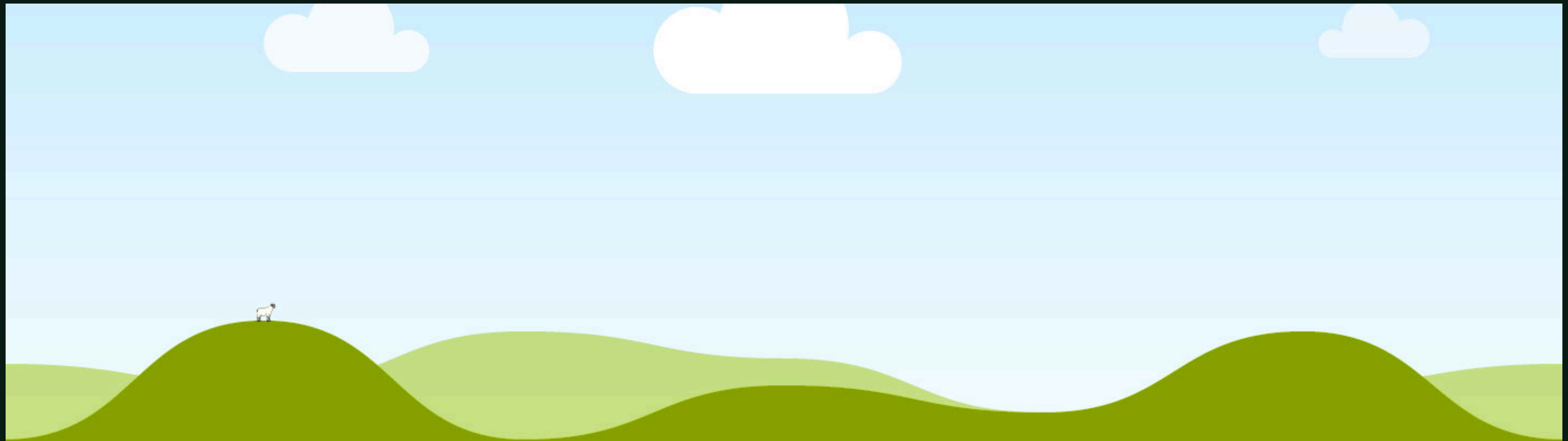
```
=====
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RM PAIRWISE ACCURACY : 0.712
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# ROOT CAUSE ANALYSIS

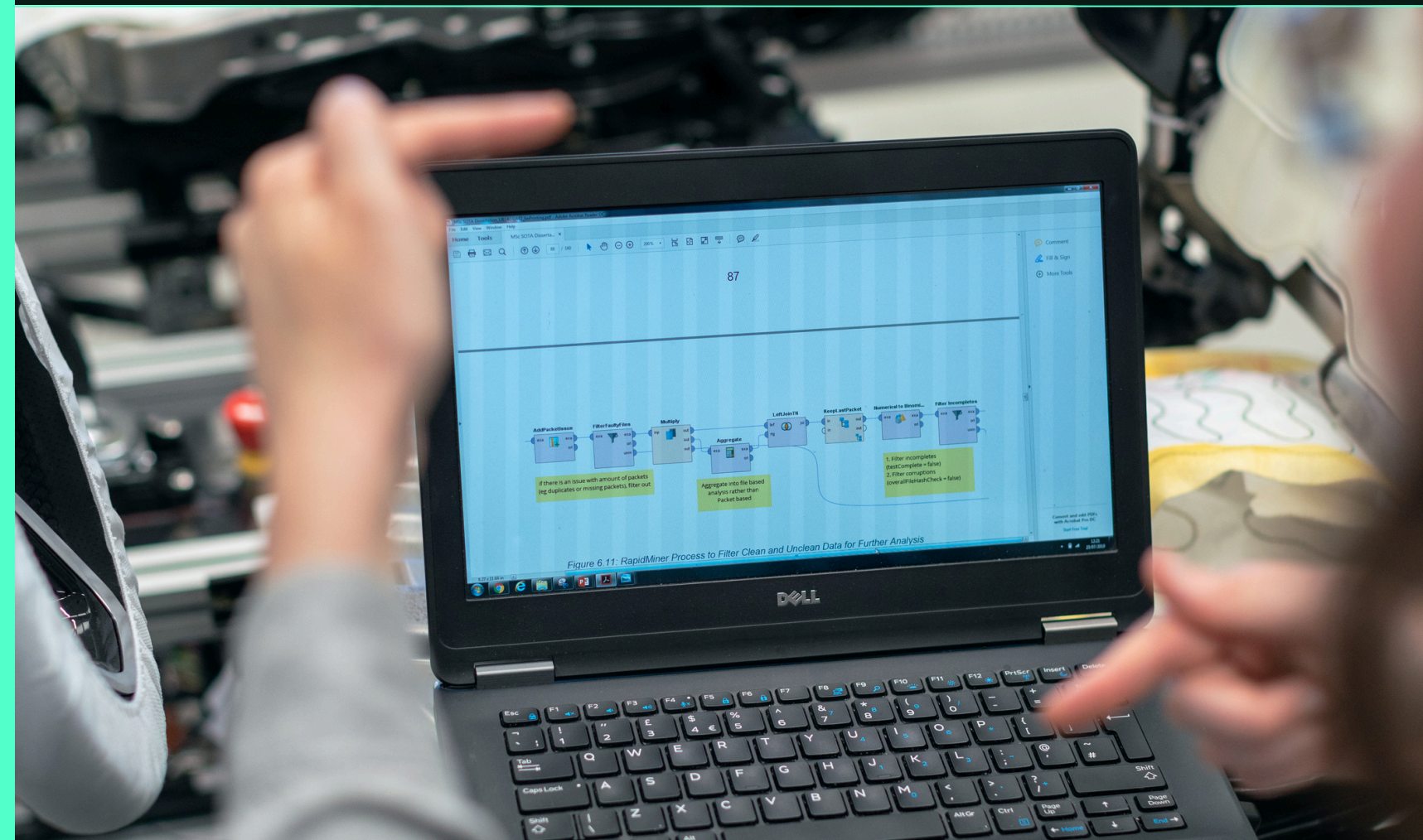
Dataset Imbalance

Can show the confusion matrix, recall, precision, f1 score





# CONCLUSION & FUTURE WORK



## From Concept to Creation

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# THANK YOU

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